Finite Semaev row-space closure audit

Independent finite-model validation

26 October 2026

Status. This is a scoped computational audit of 80 frozen Semaev draws and 200 matched-null controls, with one detailed closure witness. It does not change a Lean theorem, formal target, experiment output, official hypothesis status, or ECDLP cost claim. The executable checkers and exact machine-readable outputs are beside this PDF as `audit.py`, `closure.py`, and their JSON evidence files.

Question and sources

Does the selected degree-three row span of the fixed $n = 4, n' = 4, s = 1, x_R = 1$ Semaev S_3 system satisfy the multiplication closure needed to identify it with a $V_{F,3}$ filtration? The archived equation hash is recorded in `evidence.json`. The repository sources are `research/FFD_SEMAEV_MEASUREMENT1.md`, the corrected replay of 22 October, and the finite Lean row-space certificates. The relevant mathematical boundary is the degree-budget closure definition in Caminata–Gorla, Definition 1.5. Kosters–Yeo, Corollary 4.11 and Remark 4.12 give the trace identity and describe degree-two first falls as usual, rather than universal under every factor-base choice.

Reproduction and controls

The checker independently evaluates $S_3 = e_2^2 + e_3 + 1$ on all 256 Boolean assignments over $\mathbb{F}_2[X]/(X^4 + X + 1)$, then uses a Möbius transform to recover all four archived component equations. It confirms that $x_R = 1$ is the abscissa of curve points with $y = 6$ and $y = 7$. Boolean products use OR on variable masks and XOR on coefficients. A strict multiplier-degree budget and the historical actual-product-degree row selector yield the same rank and top-degree projection rank for this sample:

Degree	Strict rows	Product-degree rows	Rank	Projection rank
1	1	17	1	1
2	12	28	11	10
3	64	80	53	41

The archived 80-draw chart is preserved. The additive chart below labels its 68/12 corrected outputs as *direct-row grades*; their polynomial-ring first-fall interpretation remains unproved. A separate Boolean quotient calculation is summarized on page three.

Closure witness and limit

The published degree-two witness W belongs to the selected rank-53 degree-three span and lies outside the selected degree-two span. Multiplying by u_0 gives a degree-three polynomial outside that rank-53 span; adjoining it raises the rank to 54. Since $\deg W = 2$ and $\deg u_0 = 1$, the selected degree-three span is not closed under this Boolean-quotient degree-budget product. This does *not* invalidate the Lean theorem about the literal 80 rows. It shows that a separate bridge is needed before calling that span the full closed $V_{F,3}$ filtration.

Fixed Boolean quotient closure

We computed the least subspace of $\mathbb{F}_2[u_0, \ldots, u_7]/(u_i^2 - u_i)$ containing the four source-bound equations and closed under fg when $\deg g \le D - \deg f$, using squarefree monomial degree. Three methods agree on each space: degree-filtered elimination with repeated variable products, degree-filtered elimination with all allowed monomials, and a separate projection-kernel iteration. Their joint rank equals each method's rank. At degrees $1, 2, 3$, the closed ranks and top-degree projection ranks are $(1, 1), (11, 10), (79, 56)$. Thus the low-degree intersection at degree three has dimension $79 - 56 = 23$, and its new quotient count is $23 - 11 = 12$. The selected direct-row statistic instead gives $53 - 41 - 11 = 1$. The degree-two quotient count remains zero. This is one fixed draw, not the 80-draw distribution or the polynomial-ring $V_{F,D}$ with Boolean field equations. The quotient-to-polynomial-ring bridge and a Lean closure certificate remain open.

Operational limit. No heavy Lean build was started. GitHub Actions reports a payment-or-spending-limit condition before assigning runners, and the Spot host lacks the required 16 GiB reserve for a heavy local build. The complete report with source links and exact witness masks is in `report.md`.

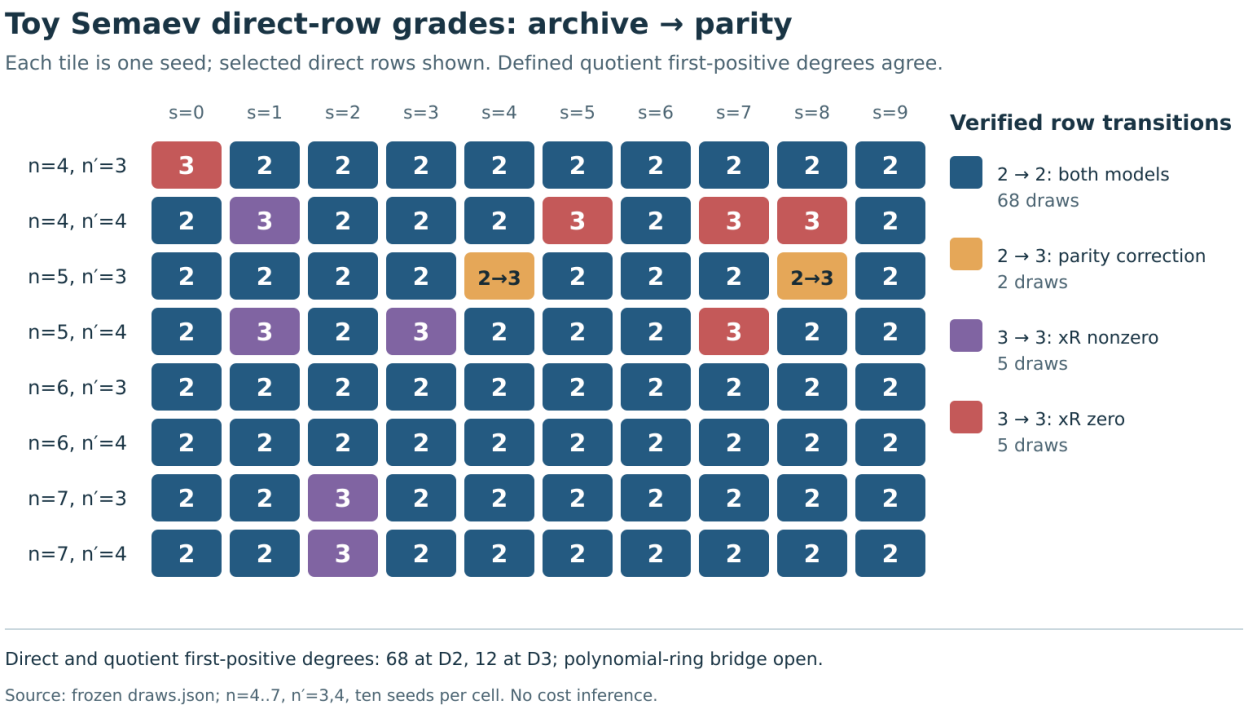


Figure 1: Archived set-product and parity-corrected direct-row grades for eight fixed cells of ten draws. The renderer verifies all 80 tiles against the archived SVG. This is neither a timing plot nor an attack-cost result.

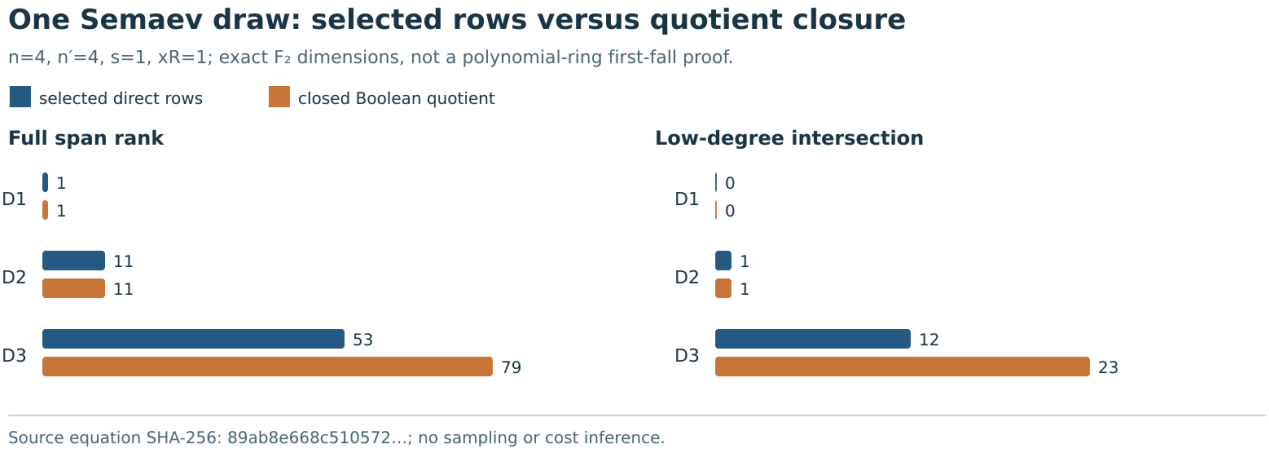


Figure 2: The selected direct rows and the full degree-bounded Boolean quotient closure for one source-bound draw. At degree three, their ranks are 53 and 79. This is no polynomial-ring or attack-cost result.

All 80 archived draws in the defined Boolean quotient

The independent checker `distribution.py` re-derives all archived component equations from field S_3, validates the field moduli, basis independence, curve target, and per-draw equation hashes, and reproduces the selected-row profiles. Three finite fixed-point methods agree on each closed quotient space at degrees one, two, and three. The first degree with a positive new-class count agrees draw by draw: 68 at degree two, 12 at degree three. This agreement is not equality of spaces or multiplicities:

Number of draws	$D = 1$	$D = 2$	$D = 3$
Selected and quotient ranks differ	2	68	74
New-class counts differ	0	23	77
Selected rows outside the quotient closure	2	5	0

The selected row rule admits products by their actual degree after Boolean cancellation; the defined closure constrains multiplier degree by $D - \deg f$. Seven selected spaces therefore contain rows absent from that closure. The exact 80-draw profiles are in `distribution-evidence.json`. Normal and optimized Python runs agree byte for byte. This is finite computational evidence, not a Lean certificate or an identification with polynomial-ring $V_{F,D}$.

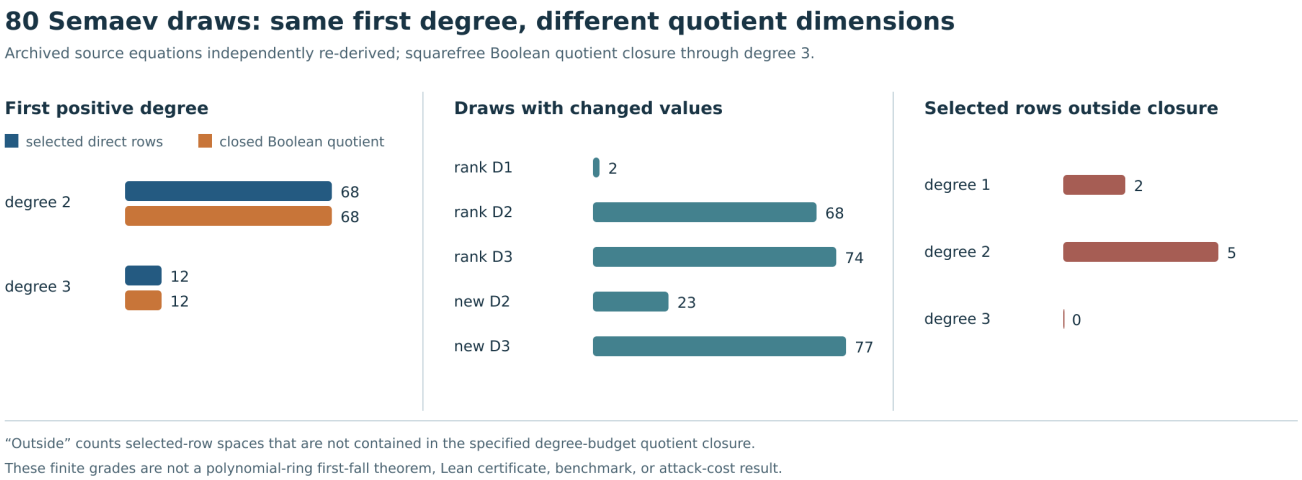


Figure 3: First-positive degrees agree, but selected and Boolean quotient dimensions usually differ. Each count is based on the same 80 frozen draws; no new sampling or performance inference is included.

The 200 archived support-matched null controls

The checker `null_distribution.py` validates each archived null's equation hash and per-equation monomial-degree multiset against its recorded Semaev source draw. It reproduces all selected-row profiles through degree six. Three methods agree on each defined Boolean quotient closure space, and the spaces are nested by degree. These are the existing controls; no new random systems were generated.

First positive degree	Selected direct rows	Closed Boolean quotient
2	13	1
3	111	117
4	76	82

Twelve nulls move from degree two to three and six from degree three to four; 182 retain their label. Selected rows are outside the specified quotient closure in 69 controls at degree two and 23 at degree three. The exact per-control profiles are in `null-distribution-evidence.json`. The 80 Semaev draws retain their 68/12 first-positive labels under this quotient, but the Semaev and null sets have different sizes. No significance, general curve, security, or performance inference is made from this finite contrast.

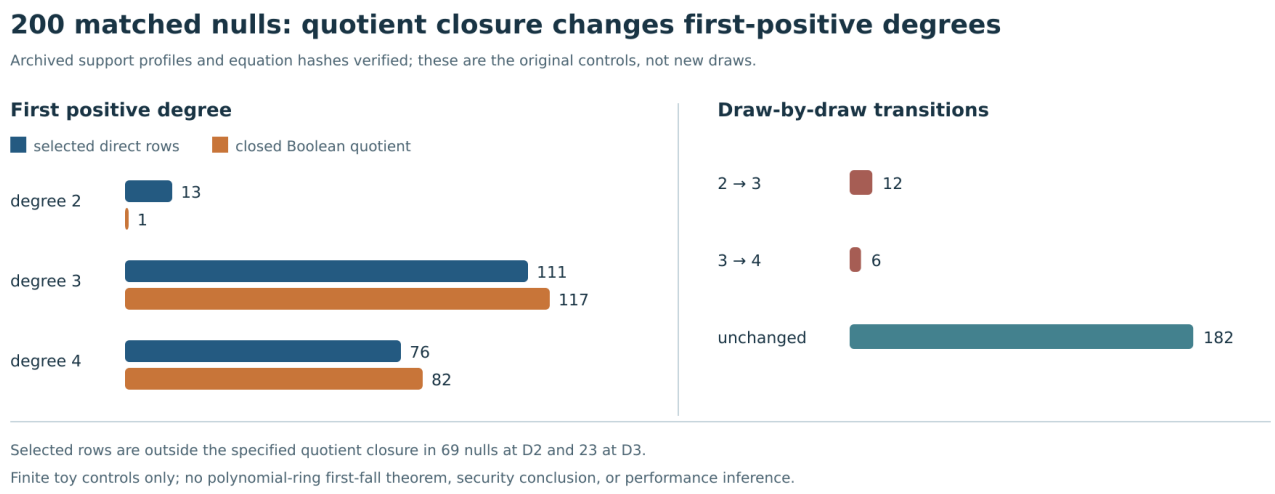


Figure 4: Selected-row versus degree-bounded Boolean quotient first-positive degrees for the 200 archived matched nulls. This is no polynomial-ring first-fall theorem or attack-cost result.